DHANUSH RAMACHANDRAN

dhanushramachandran1@gmail.com +1(646)-262-6952 Chicago, IL linkedIn github

EDUCATION

Master of Science in Applied Data Science (4.0/4.0), *University of Chicago* *08/2025 – 12/2026 | Chicago, IL, USA*

B. Tech Electrical and Electronics Engineering (9.23/10), *Vellore Institute of Technology* *08/2020 – 05/2024 | Vellore, India*

CERTIFICATES

- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect-Associate

EXPERIENCE

DATA & AI Platforms Engineer Intern, *AVANT (Fintech)* *06/2026 – 08/2026 | Chicago, IL*

- Built an end-to-end **legal-assistant chatbot** in Databricks and Slack, automating initial investigations from **100+ legal documents** for business and product teams and reducing legal research effort by **95%** with human-in-the-loop review
- Engineered a five-stage legal knowledge pipeline in **Databricks AI Search** for document ingestion, agent-ready doc-conversion, indexing, and SCD-2 type knowledge-base maintenance; built a **Supervisor–Worker architecture** grounded in current legal documents, reducing investigation time from **1 day to 10 minutes (97.9%)**
- Developed a **Triage–Investigator Agent** for Airflow failure and performance RCA, reducing human review time by **96%**; integrated Genie Spaces and **Datadog MCP** for grounded diagnostics and delivered investigations through Slack and Databricks Apps
- Built Databricks monitoring dashboards for **Airflow reliability, warehouse query performance, and cloud spend**, with Slack-based alerts
- Migrated Airflow logging and event publishing to a Listener-based plugin integrated into the Airflow image build, reducing **DAG runtime by 20%**

AI DEVELOPER - Research Assistant, *DEPARTMENT OF PEDIATRICS, UNIVERSITY OF CHICAGO* *03/2026 – Present | Chicago, IL*

- Aggregated and mapped medical research and researcher-level data from multiple ITM institutions into **RDF** for graph-based retrieval.
- Built a **Supervisor–Worker GraphRAG chatbot** for FAQ and collaborator discovery using Graphwise, Milvus, OpenAI models, and LangGraph.
- Developed a **pgvector-based retrieval approach** that improved collaborator-discovery chatbot inference time by **12×**.
- Improved chatbot efficiency through configurable research depth, few-shot query examples, agent-specific memory, and streaming updates.

CLARITY TRAVEL TECHNOLOGY SOLUTIONS (Travel Tech)

AI RESEARCH ENGINEER *03/2024 – 10/2024 | Chennai, India*

- Automated supplier-data aggregation, NLP-based mapping, and deduplication across **3M+ hotel records in MySQL**.
- Designed a **grid-based search algorithm** after analyzing retrieval bottlenecks, improving MySQL query performance by **120×**.
- Developed vector-search-based hotel recommendations in MongoDB, improving suggestion relevance by **11%**.
- Trained and deployed **CNN pipelines** for hotel-image classification and hero-image selection in the booking platform.
- Built automated web-scraping pipelines with **Selenium, Python, and Airflow** for anomaly detection and competitor analysis.
- Built an **LLM-powered web-scraping agent** enabling natural-language extraction workflows with human-in-the-loop validation.
- Fine-tuned a **DistilBERT** model for hotel-amenity mapping and deployed the classifier in production for standardized grouping in MySQL.

AI RESEARCH ENGINEER INTERN *01/2024 – 03/2024 | Chennai, India*

- Built an **NDC onboarding mapper** using LangChain and Pinecone, reducing supplier mapping time from **6 weeks to 1 week (6× faster)**.
- Built client dashboards and reports in Apache Superset with Airflow and PostgreSQL for booking KPI monitoring.
- Implemented **LLM-based sentiment analysis** for hotel reviews as a feature for a recommendation engine.

PROJECTS

Reinforcement Learning Based-Marketing Mix Model

- Trained a **PPO-based reinforcement learning agent** to optimize store-level marketing-mix decisions across **discount, feature, and display** variables under a **$10K budget constraint**.
- Estimated a **31% revenue uplift** over a **6-month** simulated evaluation period versus the baseline policy under a **$10K marketing budget.**

Review Based Product Modelling

- Built distributed NLP pipelines on **18 GB of electronic-product reviews** using **GCP Dataproc, Spark ML, and Spark NLP** for spam classification and review modeling.
- Combined **GBT and BERT** classifiers with custom **NER and Aspect-Based Sentiment Analysis (ABSA)** models to extract product- and brand-level insights from verified reviews at scale.

Delinquency Risk Modelling

- Developed an **XGBoost** delinquency-risk model and deployed the inference pipeline using **AWS SageMaker, Lambda, S3, and EventBridge**.
- Improved **ROC-AUC by 18% vs. baseline** while reducing model retraining time from **5 days to 4 hours**.

Multimodal Search & Recommendation System

- Developed a **CLIP-based multimodal retrieval system** aligning image and text embeddings in a shared representation space for semantic cross-modal search and recommendation.
- Combined multimodal similarity with **content-based and collaborative filtering** signals to generate relevant recommendations from both visual and textual information.

TECHNICAL SKILLS

AI / Machine Learning: LLMs, RAG, Agentic AI, GraphRAG, LangChain, LangGraph, Transformers, Fine-Tuning, PyTorch, TensorFlow, Scikit-learn, NLP, Deep Learning, CNNs | **Data Engineering:** Databricks, Apache Spark, PySpark, Airflow, dbt, ETL/ELT, Data Pipelines, Data Lakehouse |
Databases and Cloud: PostgreSQL, MySQL, MongoDB, Neo4j, pgvector, Milvus, Vector Search, Graph Databases, AWS, GCP, Docker, Kubernetes, MLflow, GitHub Actions, Datadog, Python, C++, SQL, FastAPI, Flask